

Shiv Shekhar

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Education

Rutgers University New Brunswick
BS Computer Science & BA Mathematics

Sept. 2023 – Jan. 2027
New Brunswick, NJ

Stanford Online Machine Learning Specialization

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, HTML, CSS, Swift
Developer Tools: VS Code, Jupyter Notebook, Git, GitHub, Jira, Kubernetes, Docker, Linux, CI/CD, AWS, Google Cloud
Frameworks: PyTorch, Flask, Django, React, Node.js, Next.js, Angular, FastAPI, Hugging Face Transformers
Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, TensorFlow, BeautifulSoup, Selenium, SQLite3, PostgreSQL

Experience

Arka (Y-Combinator)

July 2025 – Dec. 2025

Software Engineering Intern

- Led a team of 4 engineers delivering customer-facing, internal, and business-facing systems, serving as technical lead and product owner for project planning, feasibility, and implementation
- Implemented a Python ML layer (scikit-learn, Prophet, LangChain) across 4–5 integrated data sources for RFM-based customer segmentation, churn-risk scoring (~88% accuracy), and revenue forecasting within 10% of actuals, surfacing automated insight cards to drive retention and growth decisions
- Built a warehouse management system to automate packaging workflows, and co-developed a box assortment optimization algorithm that improved packaging efficiency and generated \$1M+ in annual revenue gains for clients
- Built an AI-driven packaging design tool with 2D/3D visualization, integrating Google Vertex AI with RAG over internal templates and prior designs to generate constraint-aware concepts, reducing design time by ~20%

Paramount

June 2024 – Jan. 2025

Software Engineering Intern

New York City, NY

- Collaborated on the design and implementation of data collection systems tracking 1M+ monthly user interactions across web, mobile, and connected TV platforms, enabling scalable, efficient data pipelines
- Optimized event-driven tracking by validating and debugging in-house user journey systems, improving data accuracy by 25% and enabling real-time insights for business intelligence
- Developed and maintained API integrations to synchronize tracking data across analytics platforms, reducing manual processing by 30%
- Supported Agile development sprints through sprint planning, daily standups, and technical documentation to drive iterative releases
- Prototyped user interaction flows in Figma to improve UX clarity and communicate design intent to engineering teams

Gigs Live (Y-Combinator)

Jan. 2024 – May 2024

Software Engineering Intern

- Built a real-time video pipeline using OpenCV to dynamically overlay targeted advertisements onto live stream frames, successfully validated across 3 production streams
- Developed a rule-based personalization engine matching ad content to viewers via watch history, demographics, and location signals for targeted in-stream delivery
- Improved frontend rendering performance by optimizing page elements, video previews, and carousel interactions, reducing load latency and smoothing the live streaming experience
- Automated frame-level product insertion to eliminate manual ad placement, accelerating turnaround time for sponsorship deliverables

Projects

High-Frequency Cross-Exchange Arbitrage Engine | C++, libcurl, Multithreading, JSON

- Developed a multi-threaded arbitrage engine in C++ to detect real-time price discrepancies across Binance, Coinbase, and Kraken, achieving sub-50 ms decision latency for high-frequency trading
- Implemented low-latency HTTP pipelines using libcurl to fetch bid-ask spreads in parallel, parsing JSON and applying fee-aware route analysis to compute precise net profitability

Sentiment Analysis Tracker for Financial News | Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas

- Developed a financial sentiment machine learning model by fine-tuning a RoBERTa-based NLP network on 250K+ news headlines, achieving 95% accuracy and outperforming VADER by 12%
- Performed hyperparameter tuning and cross-validation to maximize sentiment classification precision; built real-time inference infrastructure with sub-200ms latency for investor decision-making

RUHealthy | Python, Selenium, BeautifulSoup, SQLite3, Flask, JavaScript, HTML/CSS, Swift

- Engineered an automated nutrition tracking pipeline using Selenium and BeautifulSoup to scrape Rutgers dining hall data on a scheduled basis via GitHub Actions, parsing food names, serving sizes, and macronutrients into SQLite3
- Built a Flask/JavaScript web interface for students to log and analyze daily nutritional intake, with a Swift iOS prototype, adopted by students across campus following a hackathon demo